

Title: Advanced Network Intrusion Detection System Using Machine

Abstract/Project Definition:

Cybersecurity threats have increased significantly with the rapid expansion of digital infrastructure, requiring advanced defense mechanisms to protect sensitive data and critical systems. Network Intrusion Detection Systems (NIDS) serve as essential tools in identifying and mitigating unauthorized network activities. However, traditional signature-based detection methods are limited in their ability to detect new and evolving threats, including zero-day attacks. To overcome these challenges, modern intrusion detection solutions leverage machine learning techniques to enhance detection accuracy and adaptability.

This project, **Enhanced Network Intrusion Detection Using Machine Learning Techniques**, builds upon a hybrid detection framework that combines signature-based and anomaly-based methodologies to ensure comprehensive network protection. The system utilizes supervised and unsupervised learning models, trained on the UNSW-NB15 dataset, to detect a wide range of cyber threats. Initial development was focused on enhancing detection accuracy through feature engineering and ensemble learning, while further enhancements include real-time integration, advanced meta-learning approach, advanced visualization, preventive measures, and SIEM integration.

Literature Review:

Network Intrusion Detection Systems (NIDS) have become increasingly critical as cyber threats evolve in complexity and frequency. The integration of machine learning, deep learning, and bio-inspired techniques has been at the forefront of research, aimed at enhancing detection capabilities and adapting to new attack vectors.

In this context, Paper [1] by Kumar et al. presents a comprehensive review of NIDS research from 2005 to 2020, shedding light on trends, methodologies, and the continued reliance on outdated datasets like KDD '99. The authors emphasize the need for modern datasets and propose integrating bio-inspired techniques with machine learning to improve detection accuracy. Similarly, Paper [2] by Kaif et al. discusses the challenges of detecting novel attacks in complex network environments, advocating for deep learning approaches such as autoencoders and neural networks to reduce false alarms and enhance system performance. Both papers underscore the importance of updating datasets and leveraging advanced methodologies to address modern network security challenges.

In the pursuit of improving NIDS, several studies have focused on feature selection and the application of advanced machine learning models. Paper [3] by Sheen et al. explores how decision tree classifiers can be enhanced through feature selection methods like Chi-square and Information Gain, demonstrating significant improvements in classification accuracy and processing efficiency. Paper [4] by Alkasassbeh et al. evaluates the effectiveness of various machine learning classifiers, with the J48 decision tree standing out for its superior detection

rates on the KDD dataset. These studies highlight the potential of feature selection and modern classifiers in optimizing NIDS performance.

Furthermore, recent advancements have emphasized the adoption of deep learning and innovative classification techniques to bolster NIDS effectiveness. Paper [5] by Shone et al. compares classification methods for NIDS, with self-taught learning (STL) offering a balanced approach in detecting anomalies. Paper [6] by Panda et al. investigates ten machine learning algorithms on the NSL-KDD dataset, focusing on addressing challenges in detecting minority attacks through cost-sensitive classification methods. Paper [7] by Ahmad et al. provides a systematic review of machine learning and deep learning approaches, recommending the use of frameworks like LSTM-AE to tackle data imbalance and enhance detection accuracy. These studies collectively illustrate the continuous progress in NIDS and the critical need for advanced models to keep pace with emerging cyber threats.

Work Completed till Last Semester

Our project is focused on developing a hybrid intrusion detection system (NIDS) that combines both signature-based and anomaly-based detection. The system is designed to detect known and zero-day attacks with high accuracy using machine learning models.

1. **Problem Statement :-** How can we design a Network Intrusion Detection System that effectively identifies both known attacks and unknown attacks, while simultaneously improving detection accuracy, minimizing false alarms, and adapting to the ever-evolving landscape of cybersecurity threats?
2. **Dataset Used :-** We used the UNSW-NB15 dataset, which contains over 2.5 million records with 49 features covering various attack types (e.g., DoS, Exploits, Reconnaissance, Worms).
3. **Machine Learning Models Implemented :-** We trained and evaluated multiple models.
 - **Supervised Learning:** Random Forest, XGBoost
 - **Unsupervised Learning:** Isolation Forest, Autoencoder
4. **System Architecture**
 - **Data Preprocessing:** Feature selection, encoding, and scaling to improve model performance.
 - **Model Training & Deployment:** Trained and optimized models using hyperparameter tuning.
 - **Threat Detection:** Models classified network traffic as benign or malicious.
 - **Alert System:** Stored logs and alerts in a database for further analysis.
5. **Performance Evaluation**
 - Achieved high F1-score and accuracy, prioritizing detection of intrusions while minimizing false positives.
 - Used precision-recall analysis and confusion matrices for evaluation.

Expected Project Enhancements in 8th sem

Building upon the existing framework, this semester's work introduces several key enhancements to improve the system's usability, responsiveness, and effectiveness:

1. **User Interface (UI) Development** – A graphical interface for administrators to monitor real-time network activity, visualize detected threats, and interact with the system efficiently.
2. **Enhanced Meta-learning approach**- The meta-learning approach enhances intrusion detection by combining predictions from multiple base models (Random Forest, XGBoost, Isolation Forest, Autoencoder) using a meta-classifier. The focus will be on improving accuracy, reducing false positives, and increasing adaptability to evolving cyber threats
3. **Real-Time Integration** – Implementing a live network traffic capture mechanism to process and analyze incoming data in real time, enhancing the system's ability to detect and respond to threats dynamically.
4. **Enhanced Visualization** – Improving the analytical capabilities of the system through graph-based visualization of network traffic, threat patterns, and attack trends to assist in decision-making.
5. **Preventive Measures** – Incorporating intrusion prevention mechanisms that allow the system to take proactive actions, such as blocking malicious IPs, limiting suspicious activity, and notifying administrators instantly. Beyond detection, our system will now include proactive security mechanisms like Honeypot System and Firewall Automation.
6. **Log Storage and Management**- To enhance data retention and security analysis, the system now stores all detected intrusion attempts and honeypot interactions in a structured database. This enables efficient log retrieval for forensic investigations and security audits . Along with that it will help in Historical data analysis to detect patterns in attack trends.
7. **SIEM Integration :-** To enhance security intelligence and incident management, we are integrating our system with a Security Information and Event Management (SIEM) solution.

These additions aim to elevate the NIDS from a passive detection system to a more dynamic, interactive, and preventive security solution. With real-time capabilities and enhanced user interaction, the system will be more adaptable to modern cybersecurity challenges, ensuring better threat mitigation and network resilience.

Timeline:

Week 1 - 2	Phase 1: System Architecture & Database Setup
Week 3 - 4	Phase 2: Enhancing meta model Approach
Week 5 - 6	Phase 3: UI Development & Data Visualization
Week 7 - 8	Phase 4: Real-Time NIDS Integration
Week 9 - 10	Phase 5: Preventive Security Measures(Honeypot System Deployment)
Week 11 - 12	Phase 6: SIEM Integration & Log Correlation
Week 13 - 14	Phase 7: Testing, Optimization & Final Report

PO/PSO Mapping:

This project will align with the following Program Outcomes (POs) and Program Specific Outcomes (PSOs):

- **PO1: Engineering Knowledge**

Applying knowledge of machine learning, data science, and network traffic analysis to develop an intrusion detection system (IDS) that identifies threats and anomalies in network traffic.

- **PO2: Problem Analysis**

Identifying and analyzing complex security issues in network traffic by applying advanced machine learning techniques to detect anomalies and potential threats.

- **PO3: Design/Development of Solutions**

Designing and developing a robust intrusion detection system that meets the security requirements of modern networks while minimizing false positives and considering performance metrics.

- **PO4: Conduct Investigations of Complex Problems**

Utilising research methods and analysing datasets from real-world network environments to draw valid conclusions about threat detection accuracy and system performance.

- **PO5: Modern Tool Usage**

Leveraging modern machine learning algorithms, tools, and frameworks to analyse network traffic, model behaviour, and detect potential intrusions effectively.

- **PO6: Engineer and Society**

Addressing the security needs of society by developing a system that can protect critical network infrastructures and safeguard against cyber threats.

- **PO8: Ethics**

Ensuring the ethical use of data, particularly in handling sensitive network traffic, and following security best practices for data privacy and transparency.

- **PO9: Individual and Team Work**

Collaborating effectively as a team to achieve the project's goals, dividing responsibilities, and working together to develop the intrusion detection system.

- **PO10: Communication**

Effectively communicating the project's goals, methods, and results through detailed reports, presentations, and discussions with stakeholders and team members.

- **PO11: Project Management and Finance**

Applying project management principles to ensure timely completion of tasks, resource allocation, and overall efficiency in the development of the IDS.

- **PO12: Life-long Learning**

Engaging in continuous learning to stay updated with evolving network security techniques, machine learning advancements, and threat detection methods.

- **PSO1: Proficiency in AI, Data Science, Distributed Computing**

Designing and implementing an intelligent network intrusion detection system using machine learning models and network data analysis to ensure accurate threat detection and mitigation.

- **PSO2: Capability to Work in Teams and Address Industry Needs**

Collaborating as a team to create a solution that addresses the dynamic and evolving security needs of modern networks while reducing false positives and enhancing detection capabilities.

Methodology

The model developed till last sem is a hybrid intrusion detection system (NIDS) designed to effectively detect both known cyber threats and previously unseen attacks (zero-day threats). This is achieved through a combination of signature-based and anomaly-based detection techniques, leveraging the strengths of multiple machine learning models. To ensure robust detection capabilities, the system integrates supervised learning models, including Random Forest and XGBoost, which are trained on labeled attack data to recognize known intrusion patterns. In parallel, unsupervised learning techniques, such as Isolation Forest and Autoencoder, are employed to identify anomalies in network traffic that may indicate novel cyber threats. A hybrid approach is used to further refine the detection process.

In this phase , the focus will be on real-time intrusion detection, enhanced visualization, preventive security measures, and SIEM integration to improve the system's efficiency and responsiveness. The goal is to transform the NIDS into a proactive security framework capable of detecting, analyzing, and mitigating threats dynamically. It will include:-

1. Real-Time Traffic Monitoring

- The system will be upgraded to analyze live network traffic dynamically instead of relying only on pre-collected datasets.
- PyShark and Scapy can be integrated to continuously capture incoming network packets, allowing real-time classification of normal and malicious activities.
- This ensures instantaneous threat detection and response, minimizing delays in identifying cyberattacks.

2. Meta-Learning Approach for Improved Detection

- A meta-learning framework is introduced to enhance intrusion detection accuracy and adaptability.
- Predictions from multiple machine learning models (Random Forest, XGBoost, Isolation Forest, Autoencoder) will be combined using meta-classifiers such as Neural Networks (MLP), Support Vector Machine (SVM), or LightGBM.
- This multi-layered approach will improve precision, reduce false positives, and adapt better to evolving attack patterns.

3. Log Storage & Management

- A structured database (MongoDB/PostgreSQL) can be implemented to efficiently store detected intrusions, honeypot interactions, and security logs.
- Storing logs in a centralized system allows for forensic investigation, security audits, and historical attack trend analysis.
- Advanced querying and filtering mechanisms can be incorporated to allow quick retrieval and analysis of security incidents.

4. Web-Based UI & Visualization

- A web-based dashboard will be developed to provide security analysts with a visual representation of detected intrusions and network activity.
- The dashboard includes graphs, heatmaps, and timeline charts to depict attack trends, intrusion frequency, and system behavior.

- WebSockets integration enables real-time updates, ensuring that new threats are displayed immediately without manual refreshing.

5. Preventive Security Measures

- A honeypot system will be deployed to attract attackers and capture their activities, helping in understanding attack strategies and improving the detection model.
- Firewall automation can be implemented to block malicious IPs and prevent further attempts from compromised sources.
- This approach shifts the system from being purely reactive to proactive, preventing cyber threats before they escalate.

6. SIEM Integration for Advanced Threat Intelligence

- The system will be integrated with a SIEM platform to aggregate, analyze, and correlate logs from multiple sources.
- SIEM enables automated threat correlation, where detected intrusions, honeypot logs, and firewall events are analyzed together to identify complex attack patterns.
- Automated alerting and incident response mechanisms can be implemented, ensuring faster mitigation of potential threats.

Unique Contribution

Real-Time Intrusion Detection – Integrated PyShark and Scapy for live packet analysis, ensuring instant threat detection.

Meta-Learning Approach – Combining multiple models using stacking ensembles in order to improve accuracy and reduce false positives.

Advanced Visualization & UI – Developing an interactive dashboard with real-time graphs, heatmaps, and WebSocket-based live updates.

Proactive Security Measures – Deploying honeypots to capture attacker behavior and implementing automated firewall rules to block threats dynamically.

Structured Log Management – Centralized log storage in MongoDB/PostgreSQL for forensic analysis and security audits.

SIEM Integration – Establishing connection with ELK Stack/Splunk for log correlation, automated alerts, and real-time threat intelligence.

Reference

- [1] S. Kumar, S. Gupta, and S. Arora, "Research Trends in Network-Based Intrusion Detection Systems: A Review," IEEE Access, vol. 9, pp. 157761-157779, 2021, doi: 10.1109/ACCESS.2021.3129775.
- [2] M. Kaif, P. Prajwal, and V. Laxmi, "A Study on Network Intrusion Detection System," International Journal of Finance and Management Research, vol. 6, no. 3, May-June 2024, doi: 10.36948/ijfmr.2024.v06i03.20214.
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- [7] Z. Ahmad, A. Shahid Khan, C. Shiang, and F. Ahmad, "Network Intrusion Detection System: A Systematic Study of Machine Learning and Deep Learning Approaches," Transactions on Emerging